

A large, light blue watermark of the Python logo is centered in the background of the slide.

PYTHON BASICS & ENVIRONMENT SETUP

Lecture 01 | Week 01

by

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CONTENTS

- **Introduction to Python**
 - What is Python, key features, applications.
- **Working with Jupyter Notebook & IPython** (demonstration)
 - Installing/using Jupyter Notebook.
 - Running cells, Markdown vs Code cells.



INTRODUCTION TO PYTHON



WHAT IS PYTHON?

Python is an **object-oriented scripting language** that was released publicly in **1991**. It was developed by **Guido van Rossum** of the National Research Institute for Mathematics and Computer Science in **Amsterdam**.



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Python has rapidly become one of the world's most popular programming languages. It's now particularly popular for **educational and scientific computing**, and it recently surpassed the programming language **R** as the most popular **data-science programming language**.



- **Simplicity & Readability**

Python's syntax emphasizes clarity, using **indentation** instead of **braces** or **semicolons**. The Zen of Python—"Readability counts"—guides its design philosophy. This makes it easy for both beginners and professionals to read, write, and maintain code.

- **Interpreted with Dynamic Typing**

Python is an **interpreted language—code** is executed **line by line**, aiding rapid testing and debugging. Its **dynamic typing** system lets the interpreter infer variable types at runtime, adding flexibility but requiring careful handling of type-related errors.

- **Large Standard Library & Rich Ecosystem**

Python comes with a comprehensive **standard library** that helps with tasks ranging from file I/O to testing and data handling. Beyond that, the **Python Package Index** (**PyPI**) contains thousands of third-party packages—covering domains like **web development** (Django, Flask), **data science** (**NumPy**, **Pandas**), **machine learning** (**TensorFlow**)—that expand Python's capabilities exponentially.

- **Cross-Platform Portability**

Write your code once, run it anywhere—Python is platform-independent, functioning seamlessly across **Windows**, **macOS**, and **Linux**.

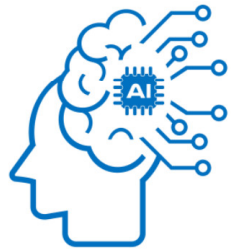
- **Multi-Paradigm Support**

Python supports **multiple programming paradigms**—including **procedural**, **object-oriented**, and **functional programming**—giving developers flexibility in how they structure code.

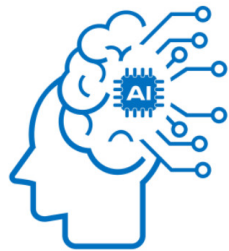
- **Garbage Collection & Memory Management**

Python includes automatic memory management, including garbage collection and reference counting, so developers don't need to manually manage memory allocation and deallocation.





1. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING



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Python remains the backbone of **AI** and **ML** development. It's widely used in **Natural Language Processing**, **Generative AI**, **Predictive Modeling**, and more. Popular libraries include **TensorFlow**, **PyTorch**, **Scikit-learn**, **Hugging Face Transformers**, and **JAX**. These tools enable the creation of everything from **Intelligent Chatbots** to **Content Generation Systems**.



2. DATA SCIENCE, REAL-TIME ANALYTICS & BIG DATA



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Python's role in **Data Science** and **Analytics** is unparalleled, with libraries like **NumPy**, **Pandas**, **Matplotlib**, powering everything from data manipulation to large-scale processing. **Jupyter Notebooks** remain a crucial tool for interactive data workflows. Python is also increasingly used in **Business Intelligence** tools and **Real-time Dashboards** via **Plotly** and **Dash**.



3. WEB DEVELOPMENT & API_s



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Python powers both robust and lightweight web applications. Frameworks like **Django** (ideal for full-featured, secure applications), **Flask** (micro-framework for simpler projects), and **FastAPI** (modern, high-performance framework with type validation and automatic API documentation) are at the forefront. They support the development of backend services, internal dashboards, and scalable APIs.

THANK YOU!